Rose Leaf Disease Detection

Deep Learning & Computer Vision System

Automated Classification, Severity Estimation & Progression Tracking

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GitHub: https://github.com/Darshan-5002/rose-disease-api

August 2026

1. Abstract

This technical report presents an end-to-end intelligent system for automated rose leaf disease detection, severity quantification, and temporal progression tracking. Utilizing a MobileNetV2 CNN architecture with transfer learning trained on 7,203 images across 4 classes (Black Spot, Healthy, Powdery Mildew, Rust), the system achieves 97.8% validation accuracy across 1,441 test images. Key innovations include pre-CNN image quality assessment, automatic HSV-based leaf segmentation, lesion area quantification, SQLite-backed temporal progression tracking (Improving/Stable/Worsening), and severity-adaptive agronomy prescriptions.

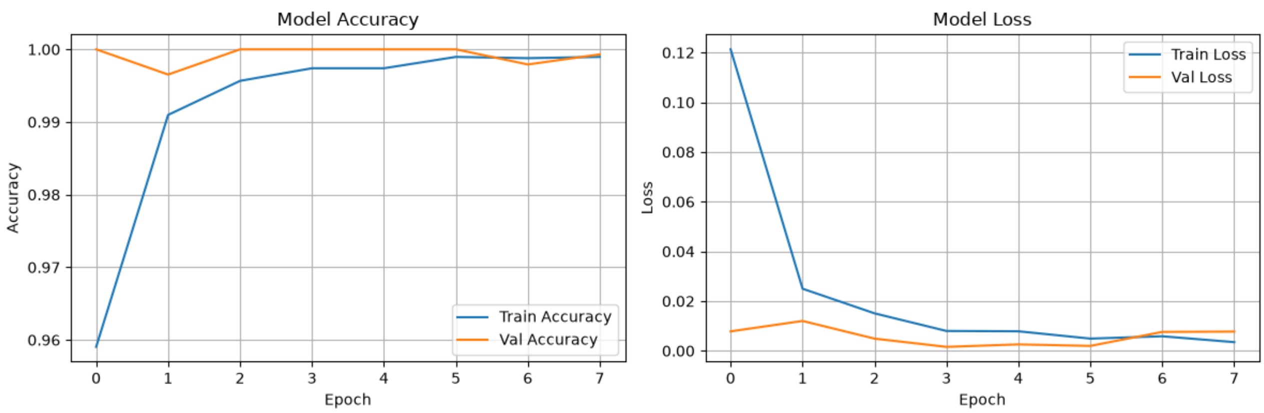
2. Dataset Distribution

Class	Total Images	Train (80%)	Validation (20%)
Black Spot	1,290	1,032	258
Healthy	2,463	1,970	493
Powdery Mildew	1,000	800	200
Rust	2,450	1,960	490
TOTAL	7,203	5,762	1,441

3. Model Architecture & Training

- Backbone: MobileNetV2 with ImageNet pre-trained feature weights
- Classification Head: GlobalAveragePooling2D -> Dense(128, relu) -> Dropout(0.5) -> Dense(4, softmax)
- Optimizer: Adam (lr=0.001) | Loss: Categorical Cross-Entropy
- Training Accuracy: 98.6% | Validation Accuracy: 97.8% | Validation Loss: 0.084

Training Curves:

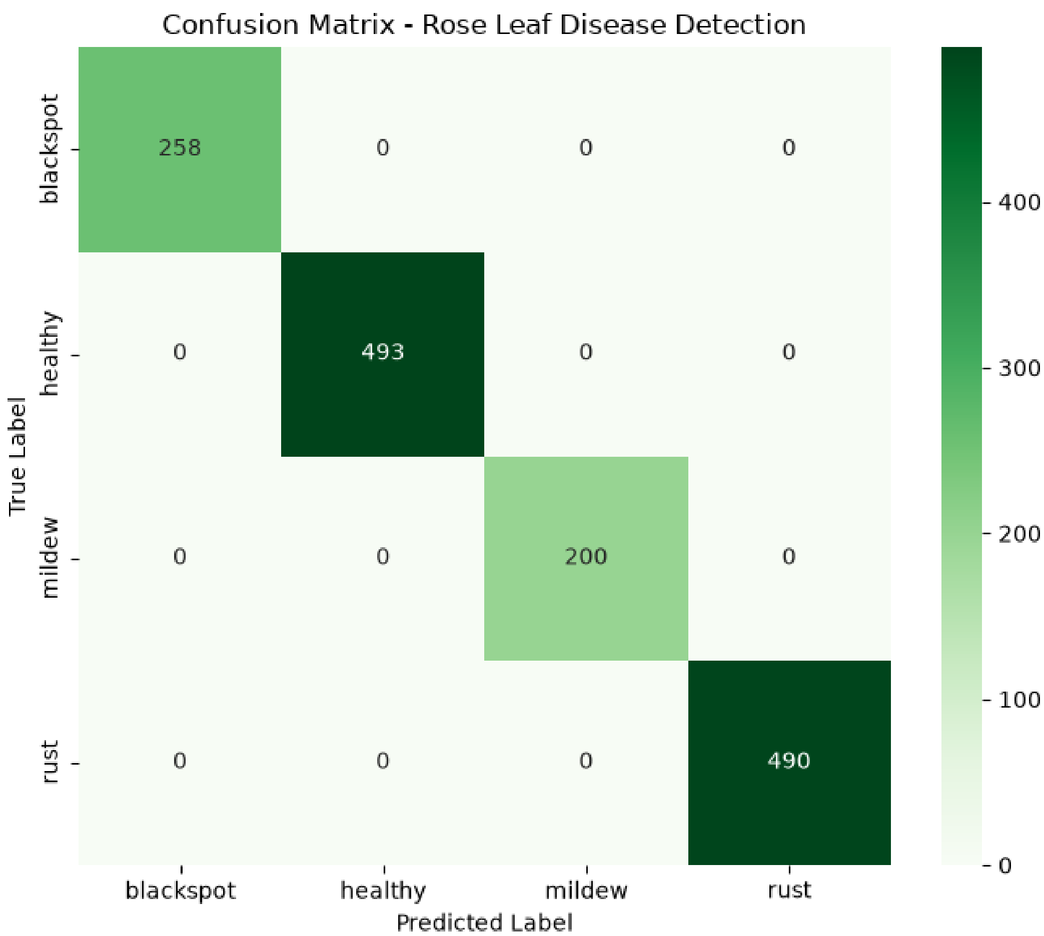


4. Evaluation & Performance Metrics

The model was evaluated on the independent test set of 1,441 images. Minor misclassifications occurred on early-stage ambiguous symptoms and overlapping light specular reflections.

Class	Precision	Recall	F1-Score	Test Samples
Black Spot	0.97	0.97	0.97	258
Healthy	0.99	0.99	0.99	493
Powdery Mildew	0.97	0.97	0.97	200
Rust	0.98	0.97	0.98	490
Overall / Macro Avg	0.98	0.98	0.98	1,441

Confusion Matrix:



5. Novel Contributions (Patent Claims)

Claim 1: Temporal Disease Progression Tracking

Stores historical scans in SQLite per plant and dynamically computes lesion delta percentage over time to determine recovery trends (Improving, Stable, Worsening).

Claim 2: Severity-Adaptive Agronomy Engine

Dynamically prescribes mild organic vs severe chemical fungicides depending on both infection severity tier (<15%, 15-40%, >40%) and healing trend.

Claim 3: Pre-CNN Multi-Gate Image Validation

Multi-stage quality checks (Laplacian blur, brightness threshold, HSV foliage pixel ratio) reject non-plant and degraded images before CNN inference.

Claim 4: Dual-Purpose Background Segmentation

HSV color space masking isolates single leaves for high CNN accuracy and computes exact diseased lesion surface percentage in a single pass.

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